

A pre-specified phase-randomization surrogate test of a second-order symbolic statistic in stage 2 sleep EEG/EOG/EMG: a negative result and its limits

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Abstract

Background. A large symbolic Markov-order statistic need not identify higher-order dynamics of the underlying signal. We compared a second-order statistic with phase-randomized signal references during Rechtschaffen–Kales stage 2 sleep (termed N2 here).

Methods. This retrospective analysis used 38 Sleep-EDF recordings from 19 participants. Each 30-s epoch received one of four symbols from whole-night median splits of two covariance-based EEG/EOG/EMG features. A common-support likelihood-ratio statistic compared pooled first- and second-order models within stage 2 runs. Each recording had 1,999 surrogates under run-level common-phase, epochwise common-phase and independent-channel phase schemes. A decision rule frozen before production required a group-level excess under both common-phase references, combining paired nights by Bonferroni and participants by Fisher with a Monte Carlo rank-model reference.

Results. Frozen group p-values were 0.877, 0.126 and 0.115, respectively; the criterion was not met. Stored-output integrity checks passed, without a complete new EDF-to-result reproduction. Post hoc, the run-level participant mean z was -0.53 (bootstrap 95% CI -0.92 to -0.14). All six lower-tail recordings were among the 27 available before the later diagnostic plan; none of the subsequent 11 fell in either tail. Median fractions of surrogate statistics above the nominal χ^2_{36} benchmark were 90.15%, 67.41% and 81.87%; these are not calibrated false-positive rates.

Conclusions. The frozen excess criterion was not met. This does not establish absence of second-order symbolic dependence: component-test power and direct surrogate-state agreement remain unvalidated, and pooled runs can reflect first-order heterogeneity. The result concerns this four-channel representation and these phase references, not a general test of cortical dynamics.

Keywords: polysomnography; stage 2 sleep; surrogate data; phase randomization; Markov order; symbolic dynamics; negative result

1 Introduction

Many analyses of physiological recordings reduce a multichannel signal to a sequence of discrete states—sleep stages, EEG microstates or quantized features—and then ask whether the sequence has memory beyond its current state. Hypnograms have long been modeled as Markov chains [1], and tests of the Markov property have been applied to EEG microstate sequences [2]. A standard tool is the likelihood-ratio statistic that compares Markov models of consecutive orders, referred to an asymptotic χ^2 distribution [3, 4]; divided by twice the number of transitions, it equals the plug-in conditional mutual information (CMI) between the current state and the state two steps back, given the previous state, and CMI-based tests have been used to estimate Markov order [5]. Three known problems complicate the interpretation of a large value. First, a coarse-grained function of a Markov process is in general not a Markov process [6], so discretizing a continuous signal into a few states can create apparent higher-order dependence without higher-order dynamics. Second, when segments with different state occupancies are pooled, the recent past carries information about the segment and therefore about the next state. Third, plug-in information estimates are

biased upwards, and with sparse count tables and a data-dependent number of populated cells the asymptotic χ^2 reference can be poorly calibrated [7, 8]. For these reasons, tests of Markov order have been proposed that resample the symbol sequence while preserving its lower-order structure [9–11].

Surrogate data provide an empirical reference at the level of the raw signal [12–14]. Fourier phase randomization produces signals with the power spectrum of the data and otherwise random phases; applying the same random phases to all channels also preserves the cross-spectra [15]. A statistic that is unusual relative to such surrogates indicates a departure from the specified random-phase reference, whose interpretation depends on exchangeability and on the structures it preserves, and surrogate tests have been widely applied to EEG [16]. Their inferential content depends on what the surrogates preserve: a statistic that responds to properties the surrogates do not reproduce, such as non-stationarity, can be shifted in either direction for reasons unrelated to the hypothesis of interest [13, 17].

This study is part of an independent research project (Brain Model, BM-2026) that represents each 30-s epoch of a polysomnographic recording by one of four symbolic states. Earlier, unpublished project analyses of this representation, which included the recordings analyzed here (Section 2.9), reported in-sample second-order dependence of the state sequences within N2 sleep, but no improvement in out-of-run prediction and no preference for a second-order model under model-selection criteria. In-sample dependence of this kind is hard to interpret, because it can arise from the three problems above and its apparent size depends on the reference distribution (Section 3.5). Here we asked a narrower, falsification-oriented question: is the N2-localized second-order statistic larger in the data than in raw-signal phase-randomized surrogates that preserve the spectral and cross-spectral structure of the signals? We used the publicly available sleep-telemetry recordings of the Sleep-EDF Database Expanded [18–21].

The decision rule was frozen before the production run, the single full execution of the frozen analysis package. The analysis is nevertheless retrospective, its documents were time-stamped in a private project library rather than in a public registry, and the analyzed recordings had been used earlier in the project (Section 2.9). We therefore describe it as internally pre-specified rather than preregistered [22]. We report the pre-specified result first and then descriptive post hoc analyses of the frozen outputs that clarify what the test can and cannot show. A pre-specified diagnostic stage that compares the lower-order structure of observed and surrogate state sequences has been implemented but not completed; it is described as planned work.

2 Methods

Figure 1 summarizes the analysis pipeline, and Table 1 the frozen analysis contract of the production run.¹

2.1 Data and cohort

The sleep-telemetry (ST) part of the Sleep-EDF Database Expanded, version 1.0.0 [18, 19], contains 44 polysomnograms of about 9 h from 22 participants, recorded in hospital in a 1994 study of temazepam effects on sleep; participants were Caucasian men and women with mild difficulty falling asleep, otherwise healthy and without other medication. Each participant contributed one night after temazepam intake and one after placebo intake. The recordings contain EEG Fpz-Cz,

¹Project identifiers used in this paper: R2 and R3, revisions 2 and 3 of the analysis package (R3 was used for the production run); BM-A004 and BM-A011, the operator and orchestrator code of R3; BM-A027, the analysis plan for the gates that follow the pre-specified test; BM-A028, the record of the G0/G1 decision; BM-A029 to BM-A031, implementations of the G2 diagnostics; BM-A032, a positive-control diagnostic. Gates: G0, execution integrity; G1, the pre-specified test; G2, lower-order diagnostics; G3, predictive value added.

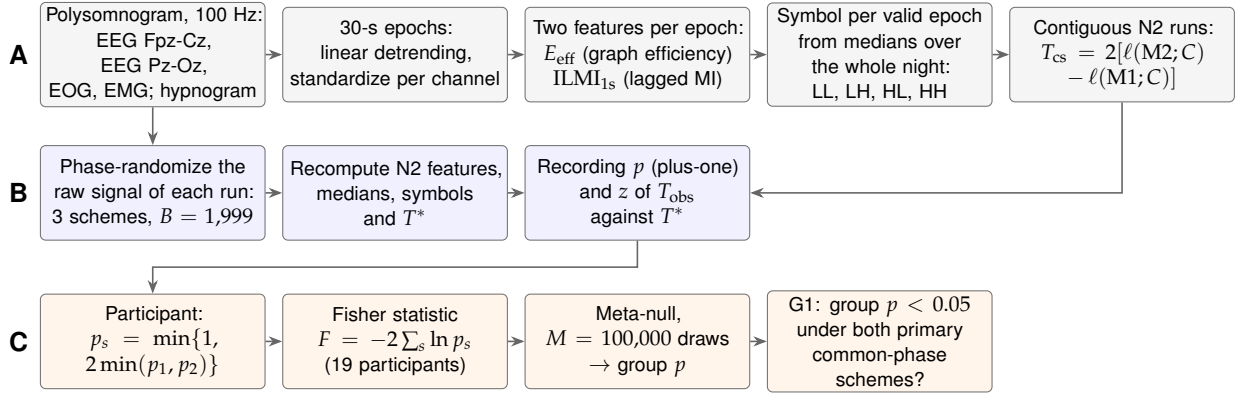


Figure 1: Analysis pipeline. (A) Observed statistic. (B) Surrogate distribution for each recording and scheme; features of non-N2 epochs keep their observed values, and the whole-night medians are recomputed in every replicate. (C) Pre-specified participant- and group-level inference and the decision rule G1. E_{eff} , weighted global efficiency; ILMI_{1s} , Gaussian lagged mutual information (MI) at 1 s; M1 and M2, first- and second-order Markov models; C, common support; T_{obs} and T^* , observed and surrogate statistic; B, surrogates per recording and scheme; p_s , participant p-value; F, Fisher statistic; M, meta-null draws.

EEG Pz-Oz, horizontal EOG and submental EMG sampled at 100 Hz, and a hypnogram scored manually in 30-s epochs by trained technicians according to Rechtschaffen and Kales (R&K) [23], based on the Fpz-Cz/Pz-Oz EEG (stages W, 1, 2, 3, 4, R, movement time and unscored). We refer to R&K stage 2 as N2 for brevity; it is not identical to stage N2 of the current scoring rules. Recordings are identified as ST7ssN, where ss is the participant number and N the night (1 or 2).

The locked analysis cohort comprised 19 participants and 38 recordings: both nights of participants 1, 2 and 4–20 (the database contains no participant 3 or 23). Participants 21, 22 and 24 had been used as pilot data in earlier stages of this project (a project designation, not one of the dataset) and were excluded before production inference. The participant is the unit of population-level inference. The temazepam and placebo nights define the paired structure of the data; this study does not test a treatment effect. The European Data Format (EDF) files were read from a mirror whose file sizes and SHA-1 digests were checked against a frozen download receipt. Per-recording values of all quantities are given in Supplementary Tables S1 and S2.

2.2 Epoch features

Each recording was cut into consecutive non-overlapping 30-s epochs (3,000 samples) from its first sample, and each epoch took the hypnogram label at its midpoint. Within each epoch, each of the four channels was linearly detrended, centered and scaled to unit standard deviation, and two features were computed. The first is the weighted global efficiency [24, 25] of the complete graph on the four channels with edge weights $w_{ij} = |r_{ij}|$ (absolute Pearson correlations) and edge lengths $1/w_{ij}$,

$$E_{\text{eff}} = \frac{1}{12} \sum_{i \neq j} \frac{1}{d_{ij}}, \quad (1)$$

where d_{ij} is the shortest-path length between channels i and j . The second, ILMI_{1s} , is the Gaussian lagged mutual information at 1 s between the four-channel vector a at time t and the vector b at time $t + 1$ s (100 samples) within the epoch [26],

$$\text{ILMI}_{1s} = \max \left\{ 0, \frac{1}{2} [\ln \det R_{aa} + \ln \det R_{bb} - \ln \det R] \right\}, \quad (2)$$

where R is the 8×8 correlation matrix of the 2,900 stacked pairs (a, b) and R_{aa} and R_{bb} are its 4×4 diagonal blocks. An epoch was valid if its label was W, 1, 2, 3, 4 or R and both features could be computed without numerical failure (finite, nonzero channel standard deviations and

Table 1: Frozen analysis contract of the production run (R3).

Item	Frozen value
Cohort	19 participants, 38 recordings; participants 21, 22 and 24 excluded
Channels	EEG Fpz-Cz, EEG Pz-Oz, horizontal EOG, submental EMG; 100 Hz
Epochs	30 s; label at the epoch midpoint; valid if W, 1, 2, 3, 4 or R and both features computable
Features, symbols	E_{eff} and ILMI_{1s} ; four symbols from whole-night medians over valid epochs
Statistic	T_{cs} on the common support of contiguous N2 runs (Eq. 3)
Surrogate schemes	run-level and epochwise common-phase (primary); independent-channel phase (sensitivity); $B = 1,999$ per recording and scheme
Seeds	$20260921 + 10007i + 100000j$; meta-null 21160921
Recording p-value	$(1 + \#\{T^* \geq T_{\text{obs}}\}) / (B + 1)$
Participant p-value	$\min\{1, 2 \min(p_1, p_2)\}$
Group p-value	Fisher statistic; meta-null with $M = 100,000$; $(1 + \#\{F^* \geq F_{\text{obs}}\}) / (M + 1)$
Decision (R3 rule 7; G1 in BM-A027)	support only if $p < 0.05$ under both primary schemes; “not supported” if $p \geq 0.05$ under both; “inconclusive” otherwise ^a
Integrity (R3 rules 8–9; G0 in BM-A027)	acceptance rate ≥ 0.99 for every recording and scheme; receipt and code hashes, checkpoint provenance, 12 production audits and package validation pass
Later changes (R3 rule 10)	no decision threshold or model definition to be changed after inspection of the full-run result; later variants are exploratory

^aCode strings of the R3 rule and of the BM-A027 claim state recorded in decision record BM-A028:

NOT_SUPPORTED_UNDER_PRESPECIFIED_COMMON_PHASE_NULLS (R3; $p \geq 0.05$ under both)

INCONCLUSIVE_NULL_SENSITIVE (R3; $p < 0.05$ under one)

RAW_SIGNAL_NOT_SUPPORTED_UNDER_PRESPECIFIED_COMMON_PHASE_NULLS (BM-A028)

positive-definite correlation matrices). Two of the four graph nodes are not EEG channels, so both features mix cortical, ocular and muscular activity.

2.3 Four-state symbolization

For each recording, the medians of E_{eff} and ILMI_{1s} over all valid epochs of the night (all stages) defined two thresholds. Each valid epoch received one of four symbols, LL, LH, HL or HH; the first letter indicates E_{eff} below (L) or at or above (H) its median, and the second letter does the same for ILMI_{1s} .

2.4 N2 runs and the common-support statistic

A run is a maximal sequence of consecutive valid N2 epochs; transitions are never counted across run boundaries. Let C be the set of positions that have two predecessors in the same run (the third and later epochs of each run), $n_C = |C|$, and x_t the symbol at position t . On C , a second-order model M2, $P(x_t | x_{t-2}, x_{t-1})$, and a first-order model M1, $P(x_t | x_{t-1})$, were fitted by maximum likelihood, pooling the runs of a recording, and

$$T_{\text{cs}} = 2 [\ell(\text{M2}; C) - \ell(\text{M1}; C)] = 2 n_C \hat{I}(X_t; X_{t-2} | X_{t-1}), \quad (3)$$

where ℓ is the maximized log-likelihood and $\hat{I}$ the plug-in CMI in nats. Both models are evaluated on the same target positions (“common support”). The nominal number of additional free parameters of M2 is $16 \cdot 3 - 4 \cdot 3 = 36$; the descriptive support-count index df_{eff} is the number of nonzero cells minus the number of nonempty context rows of the 16×4 count table, minus the same quantity for the 4×4 table. We also report $\text{CMI} = T_{\text{cs}} / (2n_C)$ and a Miller-Madow-type value $\text{CMI}_{\text{MM}} = (T_{\text{cs}} - \text{df}_{\text{eff}}) / (2n_C)$ [7], which subtracts this observed support-count index, not

validated likelihood-ratio degrees of freedom, and does not remove the small-sample bias of sparse tables; it is a descriptive quantity, not a corrected estimate.

T_{cs} replaced an earlier (“legacy”) statistic T_{v2} in which M1 was also evaluated at the first transition of each run, where M2 is undefined. The mismatch adds a run-start term to the statistic: $T_{v2} = T_{cs} + S + H$, where S is twice the negative log-likelihood of the run-start transitions under their own maximum-likelihood fit and $H \geq 0$ is a heterogeneity term. T_{v2} is retained only as a regression check.

2.5 Surrogates

For every N2 run, the raw four-channel signal of its epochs (as stored in the EDF file, without detrending or end-matching) was phase-randomized [12, 13]: the discrete Fourier transform of each channel was multiplied by $e^{i\varphi_f}$, with phases drawn uniformly from $[0, 2\pi)$ and set to zero at frequency zero and, for even lengths, at the Nyquist frequency, and transformed back. Three surrogate schemes (called “null variants” in the code, the analysis plan and Supplementary Data S1) were frozen:

- (i) **Run-level common-phase** (`common_phase_run`; primary): one random phase vector per run, shared by the four channels. It preserves the periodogram of each channel over the run and all cross-periodograms, hence the run means and the circular auto- and cross-covariances of the run [15]. The transform randomizes the nontrivial Fourier phases and disrupts phase relationships not fixed by these spectral constraints. Finite random-phase signals are not guaranteed to be exactly Gaussian or free of every higher-order dependence. The DC and Nyquist phases are held fixed; in particular, fixed Nyquist phase requires qualification of any claim of exact circular stationarity. Because the raw run was neither detrended nor end-matched, drift and the mismatch between the end points contribute broadband power to the run periodogram, which the surrogate distributes over all epochs of the run.
- (ii) **Epochwise common-phase** (`epochwise_common_phase`; primary): the same operation applied separately to each 30-s epoch of the run. It preserves the periodograms and cross-periodograms of every epoch, hence exactly the epoch mean, the zero-lag covariance matrix and the circular 1-s-lag covariance matrices of the raw epoch. These identities do not extend exactly to detrended epochs or the non-circular 1-s-lag estimate used for $ILMI_{1s}$. Both features use within-epoch covariance quantities, but exact preservation of the raw spectral quantities does not establish preservation of the features after detrending and non-circular lag estimation. State changes may occur near thresholds and may also reflect slow drift redistributed by randomization before detrending. Feature changes and per-epoch symbol agreement have not been measured on the present data. The scheme disrupts within-epoch phase organization not fixed by the periodograms, including the timing of transient waveforms, and phase continuity across epoch boundaries, which the statistic does not use. It does not guarantee removal of every higher-order dependence. The frozen analysis plan describes it as a null that preserves E_{eff} and zero-lag structure.
- (iii) **Independent-channel phase** (`independent_channel_phase`; sensitivity): an independent phase vector for each channel over the run. It preserves the autospectra and randomizes the nontrivial cross-channel phase relations on which both features depend, so it is a null of a different kind.

In each replicate, both features were recomputed for every N2 epoch from the surrogate signal, the features of non-N2 epochs and the validity mask were kept at their observed values, and the whole-night medians, the symbols and T_{cs} were recomputed exactly as for the data. A replicate was discarded if any recomputed feature could not be computed or was not finite (at most $5B$ attempts). For each recording and scheme, $B = 1,999$ accepted replicates were generated with

seed 20260921 + 10007 i + 100000 j , where $i \in \{0, \dots, 37\}$ is the position of the recording in sorted order and $j = 0, 1, 2$ indexes the run-level, independent-channel and epochwise schemes (code order). Seeds and seed formulas are written without thousands separators.

2.6 Recording-level inference

For each recording and scheme, $p = (1 + \#\{T_b^* \geq T_{\text{obs}}\}) / (B + 1)$, a one-sided test for an excess [27, 28] whose smallest attainable value is $1/2,000 = 0.0005$, and, descriptively, $z = (T_{\text{obs}} - \bar{T}^*) / \text{sd}(T^*)$, where $\bar{T}^*$ is the surrogate mean.

2.7 Participant- and group-level inference

The two nights of a participant were combined by the Bonferroni rule $p_s = \min\{1, 2 \min(p_1, p_2)\}$, and the participants by Fisher’s statistic $F = -2 \sum_s \ln p_s$ [29]. Fisher’s method combines one-sided p-values and tests the global null hypothesis that no participant deviates in the direction of an excess [30, 31]; it is not a test of an effect that is consistent across participants [32], and statistics that lie below their surrogate distributions can only increase the combined p-value. Because p_s is discrete and conservative—under the independent, untied nightly-rank model $P(p_s = 1) \approx 1/4$ —the χ_{38}^2 reference is not exact. It is nevertheless a valid conservative reference if nightly p-values are super-uniform and participants are independent, regardless of dependence between paired nights: Bonferroni makes p_s super-uniform, so $-2 \log p_s$ is stochastically bounded above by χ_2^2 , and the independent sum by χ_{38}^2 [33]. This does not establish validity of the raw-signal p-values. The null distribution of F was obtained instead by a Monte Carlo meta-null with $M = 100,000$ draws (seed 21160921 = 20260921 + 900000; one random stream for the three schemes, processed in code order). In each draw one surrogate value t was selected at random for every recording, its leave-one-out p-value $\#\{T_b^* \geq t\} / B$ was computed against the full ensemble (t counts itself, so this is a plus-one p-value against the remaining $B - 1$ surrogates), and p_s and F were formed as for the data. The group p-value was

$$p_{\text{group}} = \frac{1 + \#\{F^* \geq F_{\text{obs}}\}}{M + 1}. \quad (4)$$

An earlier version of the analysis package (R2) added a second +1 to the leave-one-out rank; an internal audit found the error before the production run, and the production package (R3) corrects it.

If the surrogate values of a recording are untied, the rank of a randomly selected surrogate is uniform on $\{1, \dots, B\}$ whatever the values are. With independently sampled ranks between nights and participants, the meta-null is then a Monte Carlo approximation of a known discrete distribution, identical for the three schemes. Independent surrogate seeds do not establish independence of the observed nightly p-values. We computed the rank-model distribution with FFT rounding bounds [31] (Supplementary Methods S1) and report exact group p-values next to the frozen Monte Carlo values.

2.8 Decision rule

The rule was frozen as rule 7 of the decision rules of the R3 analysis package before the production run (Table 1; timeline in Table 2). Support required a group p-value below 0.05 for both the run-level and the epochwise common-phase schemes (an intersection–union rule [34]); $p \geq 0.05$ for both was to be reported as “not supported under the pre-specified common-phase nulls” and disagreement as “inconclusive (null-sensitive)”. The independent-channel scheme was a sensitivity analysis. Interpretation was conditional on execution integrity (R3 rules 8–9), later formalized as gate G0 in BM-A027 (Section 2.10), and rule 10 declares any analysis variant defined

after inspection of the full-run result exploratory. For valid component tests, an intersection–union rule requires no multiplicity adjustment across its two components. Under the specified independent, untied nightly-rank model its size is at most 0.051 with the frozen Monte Carlo critical values (Section 3.2), but its power cannot exceed that of the less powerful of the two component tests. In the project’s later analysis plan (BM-A027) the rule is gate G1, followed by a diagnostic gate G2 and a predictive value-added gate G3 (Section 2.11).

2.9 Pre-specification status and prior contact with the data

Table 2 lists the relevant timestamps. They come from file metadata of the private project library (Google Drive), from times recorded inside the documents and from Git commits; none of them is a public registry entry. The 38 analysis recordings had been used before the production run in five ways. (i) Earlier in-sample analyses of the four-state representation (Introduction) included these recordings, so their observed in-sample statistics were known. (ii) The restriction of the statistic to N2 was introduced post hoc in those analyses (the operator’s documentation refers to a “post-hoc N2 localization”), so the choice of N2 is not independent of the analysis recordings. (iii) The legacy statistic of recording ST7022 had been computed and serves as a frozen regression constant (audit 06). (iv) Earlier package versions were smoke-tested on ST7011 and ST7012 with $B = 1$ on the day the production package was frozen; whether these outputs were inspected is not recorded. (v) BM-A027 was frozen on 29 September 2026 while the production run was in progress, when the recording-level outputs of 27 of the 38 recordings existed but no group-level result had been computed; whether those production outputs were inspected is not recorded. BM-A027 inherits the G1 rule unchanged, and it states the purpose of G2 as determining “whether the raw-null shift in T_{cs} is tightly coupled to simpler changes in K4 [four-state] occupancy or first-order transition structure”; G2 should therefore be read as a pre-specified examination of a phenomenon that may already have been noticed, not as a blind test. Some project files call BM-A027 a “preregistration” (for example, its file name); it is an internal, time-stamped pre-specification, not a registry entry.

2.10 Execution integrity (G0)

G0, defined in BM-A027 and distinct from the additional R3 package-validation requirement, required 38 of 38 complete recording checkpoints with exactly matching provenance; $B = 1,999$ accepted replicates and an acceptance rate of at least 0.99 for every recording and scheme; the frozen seeds and meta-null size; participant-level inference over 19 participants with the frozen two-night handling; all 12 production audits passing; and a consistent group summary. R3 rule 9 also requires package validation. The production log records runner hash checks; a fresh invocation of the recovered frozen package validator passed all 10 core and 12 payload checks. All 38 stored checkpoints also passed their original provenance validator, and independent numerical checks found all 114 surrogate ensembles finite and nonnegative, with 1,999 values each. Recomputed recording summaries agreed within 2.85×10^{-14} ; reproducing the frozen 100,000-draw meta-null yielded the archived group p-values. These checks verify stored execution evidence, not a new complete raw-signal transformation. The archived BM_38 report records ST7011 replay of two replicates under each of three schemes with maximal $|\Delta T_{cs}| = 0$; this is historical partial replay evidence, not a complete independent reproduction. The 12 audits are listed with their verbatim names in Supplementary Table S3. Three of them are weaker than their names suggest, and a fourth checks only a schema. The spectral audits (07, “autospectrum preservation”, and 08, “common-phase cross-spectrum preservation”) check, for one run per recording and scheme (the first run of the first attempt), that the largest absolute deviation of the surrogate periodograms or cross-periodograms, divided by the largest periodogram value, is below 10^{-10} ; they verify the implementation of the transform, not the preservation of the derived features. Audit 11 (“finite-ensemble pseudo-rank calibration”) is a unit test of a re-

Table 2: Timeline of the analysis (UTC). Sources: D, file metadata of the private project library; I, time recorded inside the document; G, Git commit date.

Time (UTC)	Event	Source
2026-09-26 09:16, 10:22	Smoke tests of earlier package versions (BM-A005, BM-A009) on ST7011 and ST7012 ($B = 1$)	D
2026-09-26 10:33	Internal audit finds the leave-one-out rank error of package R2; R2 put on hold	I
2026-09-26 10:40–10:47	Correction recorded (10:40); R3 package with the frozen decision rules stored (10:47; SHA-256 9ab675e3...)	I, D
2026-09-26 19:38	Production run R3 starts (first of several sessions)	D, I
2026-09-29 19:54	BM-A027 frozen (file stored 19:58, D); 27 of 38 recordings complete, no group result	I, D
2026-09-29 20:01	Amendment-001 (redefinition of G3)	I, D
2026-09-29 20:36	Launcher BM-A028 first used (run-stream log created); further sessions followed	D
2026-10-02 03:34	Recording-level summary written	D
2026-10-02 04:04	Group-level summary and audit table written; run ends	D, I
2026-10-02 04:05–06:14	G0/G1 decision record BM-A028 created; post-run checkpoint audit; final version 06:14	D, I
2026-10-02 07:25, 07:32	G2 smoke tests of the superseded implementations BM-A029 and BM-A030 (ST7011, $B = 2$; identical values)	D, I
2026-10-02 08:19	Only completed job of a full-mode BM-A029 run written (ST7011, run-level, $B = 1,999$)	D
2026-10-02 16:50:00	Amendment-002 (G2 state order and undefined transition rows)	I
2026-10-02 16:50:31	BM-A027 and Amendment-001 first committed to the repository (8b5b004)	G
2026-10-02 17:16	Amendment-002 first committed (8609c1f)	G
2026-10-02 19:04	G2 runner BM-A031 Rev2 first committed (f34d37a)	G
2026-10-03	BM-A031 Rev2 audited and promoted to canonical (311d0e5); full G2 not completed	I, G

implementation of the rank formula on the array $1, \dots, 10$, not a calibration of the meta-null, and audit 12 (“group calibration schema”) checks the columns, schemes, statistic name, number of participants, B , M and the range of the group p-values. The archived BM-A028 launcher notebook at historical commit 89ad55e records the 29 September session: package validation and `META_NULL_CALIBRATION_TEST.py -N 300000` were called with `check=True`, followed by a successful dry-run marker. This supports successful calibration in that restart session; the numerical value was not captured and it does not establish the first BM-A024 pre-flight execution. A new default run ($B = 1,999$, $N = 300,000$, seed 20260926) gave 0.0494 within the pass band 0.047–0.053. This checks the independent-night rank model, not physiological calibration. The size of the combination rule under the independent rank model is reported in Section 3.2.

2.11 Planned diagnostic stage (G2), not completed

BM-A027 specifies a diagnostic gate G2 that cannot change the G1 decision: for every recording and scheme, three lower-order mismatch measures between the observed and the surrogate symbol sequences—the occupancy median-reference discrepancy D_{occ} , the occupancy-weighted self-transition mismatch D_{self} and the occupancy-weighted first-order transition mismatch D_{M1} —and their Spearman correlations with z across recordings, with participant-cluster bootstrap intervals (10,000 resamples, seed 20260929) and leave-one-participant-out analyses. G2 has not been completed and is not part of this analysis. On 2 October, after the G1 outcome, the superseded implementation BM-A029 was smoke-tested on ST7011 ($B = 2$; repeated at 07:32 UTC with its

checkpointing successor BM-A030, with identical values), and a full-mode BM-A029 run was started that ended after its first recording $\times$ scheme job (ST7011, run-level scheme, $B = 1,999$; checkpoint with D_{occ} , D_{self} and D_{M1} written at 08:19 UTC); whether its values were inspected is not recorded. A code review found that the outputs of BM-A029 and of its checkpointing successor BM-A030 mislabeled the four states and set transition rows without outgoing transitions to zero. Amendment-002 (16:50 UTC) fixed the state order and the treatment of such rows, and the corrected runner BM-A031 Rev2, which superseded the original BM-A031 patch and an intermediate revision, is in the supplied repository snapshot [35]. Rev2 cannot reuse the BM-A029 checkpoint, and no population-level G2 diagnostic has been computed. Gate G3 (out-of-sample comparison of continuous-domain temporal models) has not been run either.

The frozen diagnostics compare observed probabilities with componentwise median references. Such median vectors and transition rows need not sum to one; these quantities are therefore the specified median-reference discrepancies, not unconditionally total variation between probability distributions. The definitions are retained without post hoc renormalization. Occupancy uses all N2 epochs and M1 uses all run-local transitions, whereas T_{cs} uses third and later positions of each run; the common run context does not imply identical target support.

The supplied Rev2 source and its archived copy are byte-identical. Seven regression tests and the checkpoint self-test passed again; the narrow CI contract also passed locally in a clean Python 3.13.5 environment with its pinned dependencies. Further probes found that the partial-checkpoint validator accepts some invalid probability arrays and a non-finite replay-error value, and the runner can label a non-smoke subset selected with `-recordings` as `G2_COMPLETE`. These are safeguard limitations, not evidence that the frozen G1 data are corrupt. A separately versioned correction and validation of the complete locked inventory are required before interpreting any output as full G2.

2.12 Post hoc analyses

The following analyses of the frozen outputs were not pre-specified and are descriptive:

- (a) a two-sided view of the recording-level p-values: counts in the lower tail ($p \geq 0.975$) and the upper tail ($p \leq 0.025$), whose null probabilities are $51/2,000$ and $50/2,000$ (exact for untied statistics; ties can only increase the lower-tail probability), with one-sided binomial tail probabilities in the observed direction and either-tail (multinomial) probabilities under an independent-recording rank model; the paired-night design does not establish independence of observed nightly p-values, and independence of surrogate random-number streams is a property of the reference simulation; at the participant level we count participants with at least one night in the tail (null probability $1 - (1 - 51/2,000)^2 = 0.050$ per participant, treating nights as independent); the participant-level binomial calculation also assumes independence between participants and nights; these tail probabilities are assumption-dependent descriptive calculations, not dependence-robust tests; sensitivity to tail width is shown with 5% tails;
- (b) participant mean z for each scheme, with percentile bootstrap 95% confidence intervals (10,000 resamples of participants; seed 20261008) [36], t intervals, two-sided sign tests, exact Wilcoxon signed-rank tests [37] and leave-one-participant-out ranges;
- (c) Spearman correlations (ρ_s) of recording z -scores between schemes, with participant-cluster percentile bootstrap intervals (10,000 resamples; seed 20261007);
- (d) comparison of the surrogate mean $\bar{T}^*$ with T_{obs} ;
- (e) comparison of $\bar{T}^*$ and T_{obs} with the 95th percentile of χ^2_{36} (51.00), the asymptotic reference with the nominal and largest possible number of degrees of freedom; the corresponding support-count indices of the surrogate sequences were not stored;

- (f) T_{cs} recomputed with medians taken over valid N2 epochs only (“N2-only thresholds”; stored for every recording in the frozen run summary; no surrogate distribution exists with these thresholds);
- (g) the legacy statistic T_{v2} relative to T_{cs} ;
- (h) temazepam–placebo differences of z ;
- (i) fractions of stored surrogate replicates strictly above $\chi^2_{36,0.95} = 50.99846016571065$, their median across 38 recordings and counts with a fraction above one half (Table 7; Supplementary Data S3 and Code S2); the nominal benchmark is descriptive, not a calibrated order test;
- (j) chronological 27/11 recording split and equal-night rank-model sensitivity cases; these were added after review and are exploratory.

No multiplicity correction was applied. The p-values from these analyses are descriptive, and several of them summarize the same data rather than providing independent evidence. Intervals are written “ a to b ” when they include negative values and “ a – b ” otherwise.

2.13 Software and execution

The production package pins NumPy 2.3.5 [38], pandas 2.2.3 [39], SciPy 1.17.0 [40], MNE-Python 1.11.0 [41] and requests 2.32.5, which its Google Colab notebook installs. Final environment records, written at about 06:10 UTC on 2 October in a post-run verification session that reused completed checkpoints, report Python 3.13.15 and the pinned versions for that session. They do not establish the environment of every earlier computing session. The run was executed in several Colab sessions (launchers BM-A024 and, from 29 September 20:36 UTC, BM-A028); completed recording checkpoints were reused only if their provenance contract matched exactly (audit 09). The recorded start time (26 September 2026, 19:38 UTC) is that of the first session, and the run ended on 2 October 2026 at 04:04 UTC. The post hoc analyses were repeated with Python 3.12.14, NumPy 2.3.5, pandas 2.2.3, SciPy 1.17.0 and Matplotlib 3.10.8 [42]; all eight numerical table fragments were byte-identical to those supplied with version 2.0. The earlier supplement reports a different post hoc runtime; the present rerun establishes the environment stated here, not every earlier runtime. Separately, the repository CI contract passed locally with exactly Python 3.13.5 and its pinned dependencies; this narrow contract certifies the frozen Rev2 identity and safeguards tested by that contract, not full-project reproducibility, physiological inference, or remote GitHub Actions status. Supplementary Code S1 reproduces the baseline S1/S2-derived analyses; Code S3 reproduces the exploratory 27/11 partition and equal-night rank-model sensitivity case from S1 and summarizes Data S3; Table 7 uses separate Supplementary Code S2 and Data S3 and requires the stored ensembles, with the additional recovered R3 calibration script checked separately (Section 2.10), while Code S1 re-verifies the pre-specified group-level results (Fisher statistics, participant p-values and the exact meta-null); audit values and timestamps are quoted from the archived audit table and the provenance records.

3 Results

3.1 Data and execution integrity

The 38 recordings had a median of 960 valid epochs (range 816–1,129) and a median of 437.5 N2 epochs (201–625) in a median of 40 runs (21–82). In total, 16,761 N2 epochs formed 1,572 runs, 417 of which consisted of a single epoch; the per-recording mean run length had a median of 10.4 epochs (5.0–22.9). The common support contained a median of 360.5 positions (132–553), or 83% of the N2 epochs (66–92%). df_{eff} had a median of 34 (19–36) and equaled 36 in 7 recordings.

Table 3: Pre-specified group-level results (frozen R3 outputs) with rank-model p-values.

Surrogate scheme	Role	F_{obs}	Mean ^a	95th pct. ^a	p (MC) ^b	p (exact) ^c	p (χ^2_{38}) ^d	Mean z ^e
run-level	primary	22.86	33.10	48.97	0.8773	0.8768	0.975	-0.53
epochwise	primary	43.57	33.08	48.90	0.1264	0.1271	0.246	0.18
independent-channel	sensitivity	44.12	33.07	48.93	0.1152	0.1164	0.229	-0.17

^a Mean and 95th percentile of the Monte Carlo meta-null ($M = 100,000$); exact values 33.09 and 48.98.

^b Frozen empirical group p-value; the decision rule uses this value.

^c Independent, untied rank-model reference (Supplementary Methods S1).

^d Conservative Fisher reference for valid Bonferroni-combined p-values and independent participants; not used by the frozen decision rule.

^e Mean over participants of the mean z of the two nights.

All 12 audits were reported as passing in the archived audit table (Supplementary Table S3). The largest relative error of the surrogate periodograms was 1.04×10^{-15} and that of the common-phase cross-periodograms 9.98×10^{-16} ; no replicate was discarded (minimum acceptance rate 1.000). The recovered BM-A028 decision classified G0 as PASS. Fresh package validation, 38/38 stored-checkpoint validation, the 12 archived production audits and reproduction of the group calculation support that execution-integrity decision. The raw EDF transformation was not rerun.

3.2 Pre-specified group-level test (G1)

The observed Fisher statistics were 22.86 under the run-level scheme, 43.57 under the epochwise scheme and 44.12 under the independent-channel scheme, with frozen Monte Carlo group p-values of 0.8773, 0.1264 and 0.1152 (Table 3, Fig. 2). The exact p-values, 0.8768, 0.1271 and 0.1164, differ from the Monte Carlo values by 0.5, 0.6 and 1.1 Monte Carlo standard errors (each about 0.001); on the grid $k/2,000$ of the observed plus-one p-values they agree with these to within 1×10^{-5} . The exact null distribution has a mean of 33.09 and a 95th percentile of 48.98, below the χ^2_{38} reference (mean 38). Evaluated against the exact distribution of the observed statistic, each component test of the decision rule has a size of 0.0500 with the exact critical value (48.98), and 0.0501 (run-level) and 0.0508 (epochwise) with the frozen Monte Carlo critical values (48.97 and 48.90); the intersection-union rule therefore has a size of at most 0.051. These sizes refer to untied, uniformly distributed ranks independent between nights and participants. Valid marginal recording-level p-values alone do not guarantee this reference after the within-participant Bonferroni combination. Independent surrogate seeds define the simulated reference ensemble but do not establish independence of observed nightly p-values. The calculation audits the combination step under this rank model, not the type I error of the entire physiological-data pipeline under arbitrary dependence between nights; validity of raw-signal surrogate p-values requires additional exchangeability assumptions.

Both primary p-values exceeded 0.05, so G1 failed: the outcome under the frozen R3 rule is “not supported under the pre-specified common-phase nulls”, and decision record BM-A028 records the corresponding claim state of BM-A027 (code strings in Table 1). No participant had $p_s < 0.05$ under the run-level scheme (smallest 0.071); two participants did under the epochwise scheme (smallest 0.012) and two under the independent-channel scheme (smallest 0.001).

As a post hoc dependence sensitivity case, uniform nightly ranks on $k/2000$ with equal ranks within each pair and independent participants give group tails of approximately 0.275, 0.0033 and 0.0029. The continuous χ^2_{38} conservative upper references are 0.975, 0.246 and 0.229. The lower values concern uniform rank marginals, not all merely super-uniform nightly p-values; they do not prove robustness of the EEG pipeline under arbitrary dependence. Different dependence assumptions can change which component crosses 0.05 and therefore its descriptive category, but do not change the archived decision made with the frozen reference. Correlations of observed z-scores are not an identification of the null copula.

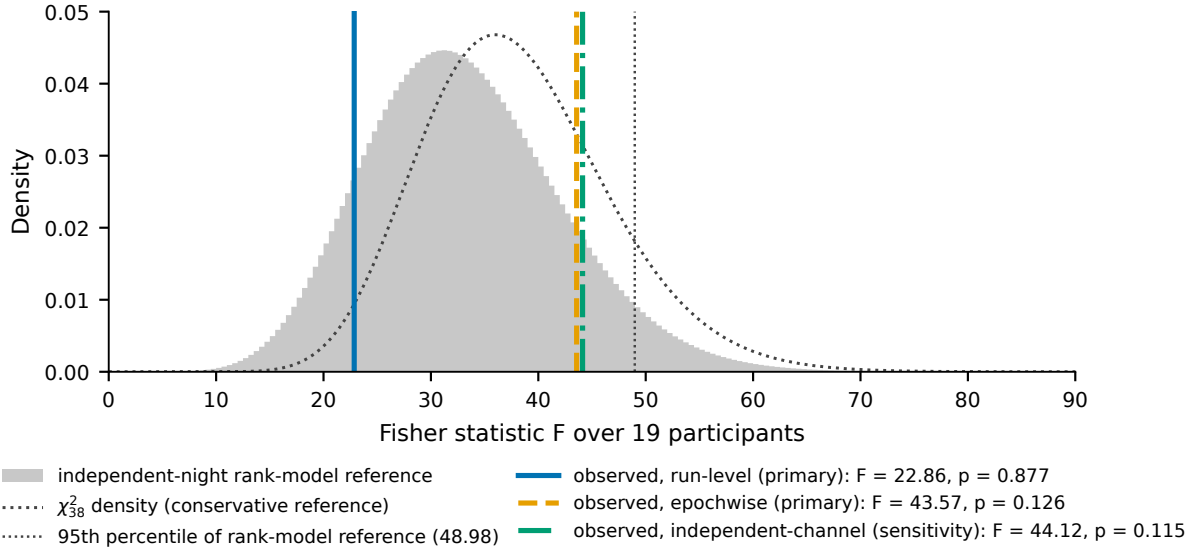


Figure 2: Pre-specified group-level test. Gray: independent, untied rank-model reference distribution of the Fisher statistic F over 19 participants (bins of width 0.5), which is the same for the three surrogate schemes; dotted curve: χ^2_{38} density, which is conservative, rather than exact, for valid Bonferroni-combined participant p -values and independent participants; dotted vertical line: 95th percentile of the exact null (48.98). Colored vertical lines: observed statistics (solid, run-level; dashed, epochwise; dash-dot, independent-channel) with their frozen Monte Carlo p -values.

3.3 Recording-level results: a two-sided view (post hoc)

The run-level lower-tail pattern was concentrated in already available outputs: the first 27 recordings (ST7011–ST7151) contained all six recordings with $p \geq 0.975$, with mean recording-level $z = -0.76$. The later 11 had no recordings in either 2.5% tail and mean $z = 0.05$. This chronological split is post hoc, is not an independent holdout comparison and does not isolate participant or treatment effects. The diagnostic plan already referred to a “raw-null shift”, so this pattern is not a prospective discovery.

Under the run-level scheme, 3 of 38 recordings had $p < 0.05$, but none lay in the upper 2.5% tail of its surrogate distribution, whereas 6 lay in the lower 2.5% tail ($p \geq 0.975$; 0.97 expected; Table 4, Fig. 3). If recordings are independent, the one-sided probability of 6 or more is 0.0004 (0.0007 for either tail); with 5% tails the counts were 9 lower against 3 upper (one-sided probability of 9 or more, 9×10^{-5}). The six recordings came from five participants (participant-level probability 0.002 under the independent-night and independent-participant rank model). The surrogate mean exceeded the observed T_{cs} in 25 of 38 recordings, the mean z was -0.53 , and the p -values were skewed toward 1 (11 recordings with $p \geq 0.9$). Under the epochwise scheme the tails were balanced (2 upper, 1 lower; mean $z = 0.18$). Under the independent-channel scheme 2 recordings lay in the upper and 4 in the lower tail (one-sided probability 0.016, either tail 0.030; participant level 0.068), the mean z was -0.17 , and the p -values were U-shaped.

Seven recordings had $p < 0.05$ under at least one scheme (Supplementary Table S4), and one (ST7182) under both primary schemes. With marginal probability 99/2,000 for each test under the untied rank model, the expected count under both primary schemes is 0.093 for independent rejection events and at most 1.881 for identical rejection events. Without assumptions on their joint distribution, the general Fréchet bounds are 0 to 1.881. A positive Spearman correlation of the z -scores ($\rho_s = 0.73$) does not establish positive dependence of the extreme-tail rejection events. The z -scores of the three schemes were concordant but not interchangeable: ρ_s was 0.89 (95% CI 0.77–0.95) between the run-level and independent-channel schemes, 0.73 (0.44–0.87) between the run-level and epochwise schemes, and 0.65 (0.36–0.80) between the independent-channel and

Table 4: Distribution of recording-level results ($n = 38$; post hoc).

Scheme	$p < 0.05$	$p \leq 0.025^a$	$p \geq 0.975^a$	Median p	Mean z	$z > 0$	$\overline{T^*} > T_{\text{obs}}^b$
run-level	3/38	0/38	6/38	0.6683	-0.53	13/38	25/38
epochwise	3/38	2/38	1/38	0.4345	0.18	22/38	16/38
independent-channel	6/38	2/38	4/38	0.6765	-0.17	13/38	25/38

^a Upper and lower 2.5% tails of the surrogate distribution, defined by the rank-based p ; about 0.95 and 0.97 recordings expected in each under the null.

^b Recordings whose surrogate mean exceeds the observed statistic.

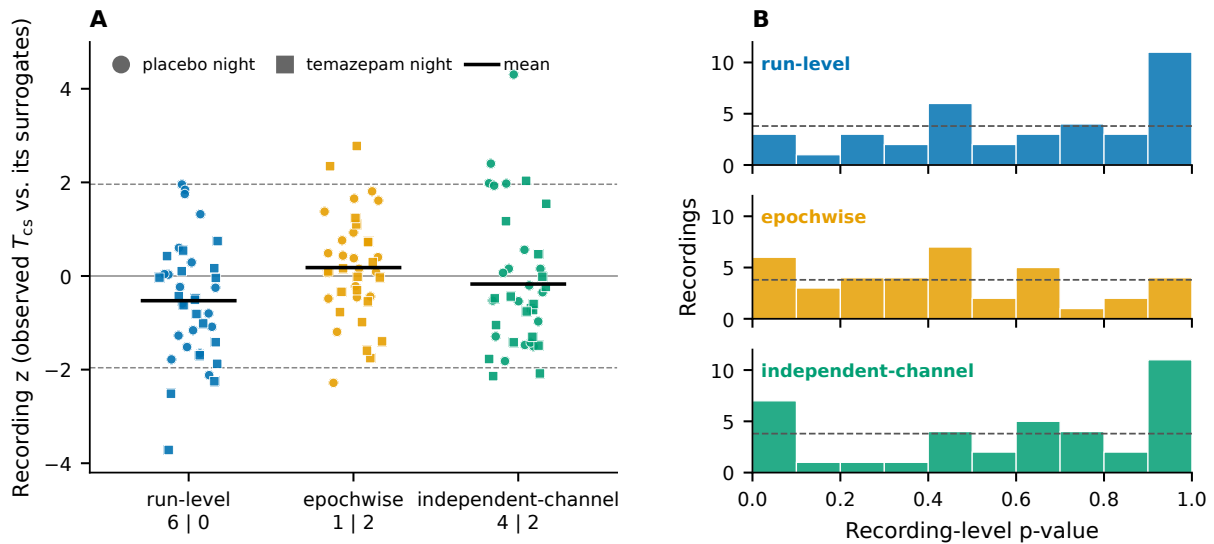


Figure 3: Recording-level results ($n = 38$ per scheme; post hoc view). (A) z-scores of the observed T_{cs} relative to its surrogate distribution; circles, placebo nights; squares, temazepam nights; thick black horizontal lines, means; thin gray line, $z = 0$; dashed lines, ± 1.96 . The numbers under each scheme give the recordings in the lower and upper 2.5% tails ($p \geq 0.975$ | $p \leq 0.025$); the tails are defined by the rank-based p , not by z . (B) Histograms of recording-level p-values (bins of width 0.1); dashed lines, the 3.8 recordings per bin expected under the null.

epochwise schemes (Supplementary Fig. S1).

3.4 Participant-level summaries (post hoc)

The participant mean z was negative under the run-level scheme (-0.53 ; bootstrap 95% CI -0.92 to -0.14 ; t interval -0.96 to -0.10 ; 5 of 19 participants positive; sign test $p = 0.064$; Wilcoxon $p = 0.018$), and omitting any one participant did not change the sign of the mean (leave-one-out range -0.61 to -0.43 ; Table 5). Under this scheme, 8 of 19 participants had $p_s = 1$, that is $p \geq 0.5$ on both nights (4.8 expected; one-sided binomial $p = 0.078$, two-sided 0.11). Under the epochwise and independent-channel schemes the participant means were 0.18 (-0.20 to 0.55) and -0.17 (-0.65 to 0.39). These summaries describe the same data and are not independent pieces of evidence.

3.5 What the surrogates preserve, and the scale of T_{cs} (post hoc)

Under the epochwise scheme, the surrogate mean of T_{cs} tracked the observed value closely across recordings ($\rho_s = 0.77$; median ratio of surrogate mean to observed value 0.98, interquartile range 0.86–1.10; Table 6, Fig. 4). Under the run-level and independent-channel schemes the agreement was weaker ($\rho_s = 0.49$ and 0.48), and the surrogate means were higher (median ratio 1.11 for

Table 5: Participant mean z ($n = 19$; post hoc).

Scheme	Mean z	95% CI ^a	$z > 0$	Sign test p	Wilcoxon p	Leave-one-out range
run-level	-0.53	[-0.92, -0.14]	5/19	0.064	0.018	[-0.61, -0.43]
epochwise	0.18	[-0.20, 0.55]	12/19	0.359	0.312	[0.09, 0.30]
independent-channel	-0.17	[-0.65, 0.39]	7/19	0.359	0.210	[-0.36, -0.07]

^a Percentile bootstrap over participants, 10,000 resamples, seed 20261008.

Table 6: Surrogate means of T_{cs} relative to the observed value and to the χ^2_{36} reference ($n = 38$; post hoc).

	$\overline{T^*}/T_{obs},$ median [IQR]	$\rho_s(\overline{T^*}, T_{obs})$	$T/36,$ ^a median [min, max]	$T > \chi^2_{36,0.95}$ ^b
run-level surrogates	1.11 [0.97, 1.40]	0.49	1.89 [1.25, 3.16]	35/38
epochwise surrogates	0.98 [0.86, 1.10]	0.77	1.57 [1.01, 2.66]	27/38
independent-channel surrogates	1.11 [0.92, 1.38]	0.48	1.75 [0.68, 2.76]	31/38
observed data	—	—	1.62 [0.79, 3.32]	26/38

^a T is the surrogate mean $\overline{T^*}$ in the first three rows and T_{obs} in the last row; 36 is the nominal and largest possible number of additional parameters of M2.

^b Recordings in which T exceeds the 95th percentile of χ^2_{36} (51.00).

both); the run-level surrogate mean exceeded the epochwise one in 28 of 38 recordings. For the same recording, the ratio of the largest to the smallest of the three surrogate means had a median of 1.25 (maximum 1.87, ST7071).

Table 7: Post hoc stored-replicate comparison with the fixed nominal $\chi^2_{36,0.95} = 50.99846$ benchmark. Each scheme has 38 recordings with 1,999 replicas each. This is not a calibrated Markov-order test or an estimate of physiological type-I error.

Scheme	Median within-recording fraction	Recordings with fraction > 0.5
Run-level common phase	90.15%	32/38
Epochwise common phase	67.41%	26/38
Independent-channel phase	81.87%	30/38

The observed T_{cs} had a median of 58.2 (interquartile range 45.0–73.7). Its ratio to the nominal 36 degrees of freedom had a median of 1.62 (range 0.79–3.32), and 26 of 38 observed values exceeded the 95th percentile of χ^2_{36} (27 of 38 exceeded that of $\chi^2_{df_{eff}}$ using each recording's descriptive support-count index; this second comparison is also uncalibrated). The surrogate means were 1.25–3.16, 1.01–2.66 and 0.68–2.76 times 36 (medians 1.89, 1.57 and 1.75 for the run-level, epochwise and independent-channel schemes) and exceeded the 95th percentile of χ^2_{36} in 35, 27 and 31 of the 38 recordings. A new post hoc calculation used the stored ensembles to count replicates strictly above the fixed nominal $\chi^2_{36,0.95}$ benchmark, 50.99846016571065 (Table 7). This benchmark is descriptive: df_{eff} is not a validated likelihood-ratio calibration, and the realized support of each surrogate sequence was not stored. Values of T_{cs} of the observed size therefore arise in run-level and independent-channel surrogates, whose nontrivial within-run Fourier phases are randomized subject to the run's periodograms (and, for the run-level scheme, its cross-periodograms), and also in epochwise surrogates, which keep the epoch-by-epoch spectral structure. These magnitudes alone do not identify within-run Markov order; no calibrated rejection rate or physiological type-I error is inferred from this comparison. The plug-in CMI had a median of 0.080 nats (0.039–0.222) and CMI_{MM} a median of 0.035 nats (–0.011 to 0.150).

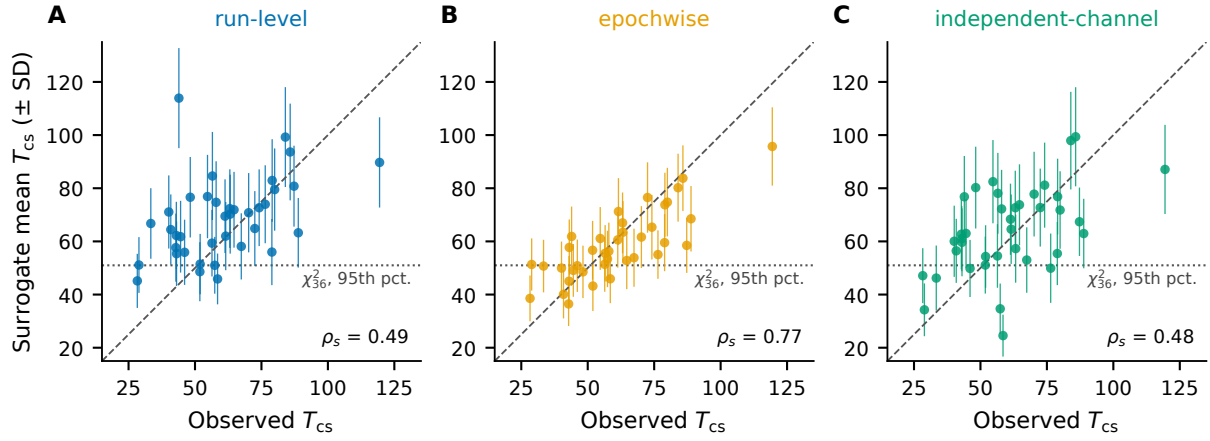


Figure 4: What the surrogates preserve (post hoc). Surrogate mean ($\pm$ SD) of T_{cs} against the observed T_{cs} for each recording under the run-level (A), epochwise (B) and independent-channel (C) schemes; dashed line, identity; dotted line, 95th percentile of χ^2_{36} (51.0); ρ_s , Spearman correlation.

3.6 Sensitivity of the observed statistic to the thresholds (post hoc)

With N2-only thresholds, T_{cs} changed by a median factor of 1.05 (interquartile range 0.94–1.20; range 0.68–1.98) and by more than 20% in 12 of 38 recordings; the largest relative increase was in ST7101 (28.9 to 57.3; $\times 1.98$) and the largest decrease in ST7172 (87.3 to 59.5), and $\rho_s = 0.79$ between the two versions (Supplementary Fig. S2). No surrogate distribution exists with these thresholds, so their effect on the test cannot be assessed. The legacy statistic T_{v2} exceeded T_{cs} by a median factor of 2.23 (range 1.64–4.93), and the excess increased with the number of N2 runs ($\rho_s = 0.85$), as expected from its run-start term.

3.7 Temazepam and placebo nights (post hoc)

The z-scores did not differ systematically between temazepam and placebo nights (Wilcoxon $p = 0.28, 0.54$ and 0.080 for the run-level, epochwise and independent-channel schemes; Supplementary Table S5). These comparisons do not test a treatment effect.

4 Discussion

4.1 Principal findings

In 19 participants, the prespecified test did not detect an excess of the N2-localized statistic T_{cs} over either primary reference, and the pre-specified test was negative. The recovered stored execution-integrity evidence supports G0 PASS; the observed Fisher statistics and participant p-values are reproduced exactly from the frozen recording-level outputs, and an exact calculation of the meta-null agrees with the frozen Monte Carlo group p-values within Monte Carlo error. Post hoc analyses (Section 3.5) further indicate that values of T_{cs} of the observed size occur routinely in surrogate signals, in particular in run-level surrogates, which contain the spectral and cross-spectral constraints of each run, run boundaries and the observed features of non-N2 epochs, with nontrivial Fourier phases randomized subject to those constraints. This does not guarantee removal of every higher-order dependence. In-sample second-order dependence of these state sequences, and in particular a significant asymptotic χ^2 test, is therefore not evidence of higher-order or nonlinear dynamics.

4.2 Why the negative result is not evidence of absence

Absence of evidence is not evidence of absence [43]; a claim of absence would require a smallest effect of interest and a test with known sensitivity to it [44]. Four features of the design limit what the failure of G1 can show.

The epochwise null conditions on each epoch's spectra. Both features are computed within single epochs from covariance quantities related to the raw spectral quantities preserved by the epochwise surrogates (Section 2.5); accordingly, the epochwise surrogate means tracked the observed values closely ($\rho_s = 0.77$; median ratio 0.98) and the two tails were balanced. This null therefore asks whether T_{cs} depends on within-epoch structure beyond the periodograms. It may retain substantial dependence of the epoch-level state sequence on its own past, although per-epoch symbol agreement has not been measured and implementation details prevent an exact invariance claim. It may therefore have limited power against some forms of second-order dependence; this is a theoretical concern that has not been quantified by a validated injection–recovery experiment. Because the intersection–union rule required both nulls to be rejected, support required excess under both references. This is a narrow alternative; it is not a general test of all forms of symbolic second-order dependence, and the relative power of the two arms need not be the same under every alternative. The negative G1 result therefore cannot establish absence of second-order dependence of the state sequence; this consequence for the power of the rule was not quantified before the run.

The run-level surrogates tended to produce larger values than the data. The observed T_{cs} lay below its run-level surrogate distribution more often than expected under the null (6 recordings in the lower and none in the upper 2.5% tail; participant mean $z = -0.53$; post hoc), which is descriptive, post hoc evidence of departure from the run-level reference under the stated rank-model assumptions. It does not identify the mechanism or establish the calibration validity of the surrogate procedure. Fisher's combination of one-sided p-values registers such a shift only as large p-values [31]. Several mechanisms could produce the shift, and we have not distinguished them. First, phase randomization can redistribute time-local phase organization across each run, so the epoch features of a surrogate fluctuate around values set by the run's average spectrum. Depending on persistence and run length, this can make within-run state sequences more or less persistent than in the data and change the occupancy of each run; pooling runs with different occupancies creates apparent second-order dependence, because the two previous states reveal which kind of run the sequence is in better than the last state alone. Second, the number of populated cells, and with it the upward bias of the plug-in statistic, can differ between data and surrogates [7, 8]. Third, the whole-night thresholds are recomputed from surrogate N2 features and observed non-N2 features, so a systematic change of the N2 features shifts the N2 occupancy. Fourth, the raw runs were randomized without detrending or end-matching, so power from drift and from the end-point mismatch is redistributed over all epochs of a surrogate, and transient N2 events such as spindles and K-complexes are dispersed; differences in stationarity between data and surrogates are a known source of shifted surrogate statistics [13, 17]. That the run-level surrogate means exceeded the epochwise ones in 28 of 38 recordings fits both stationarization and edge effects, the two properties that differ between the schemes. G2 will show whether z is associated with occupancy and first-order transition mismatch, but not which of these mechanisms is responsible. A reference distribution shifted upward relative to a specified alternative can reduce sensitivity of an upper-tail test. The relevant alternative and loss of power have not been validated here; drift, edge effects and other mechanisms remain candidates.

The sensitivity of the pipeline is unknown. No positive control has been validated. An internal sequence-level injection–recovery diagnostic (BM-A032 Phase A) was reported as failing to validate the proposed positive-control construction. Its frozen protocol, executable and Phase-A outputs have now been recovered and are indexed in the release-preparation evidence. The execution record remains REVIEW / SEQUENCE_LEVEL_DIAGNOSTIC_COMPLETE /

RAW_SIGNAL_PHASE_B_NOT_RUN. Recovery of this sequence-level diagnostic does not validate a raw-signal positive control. It is not used here to establish the power of the physiological pipeline or to infer absence of second-order structure. Without a power estimate, a non-significant group p-value cannot be turned into an upper bound on such structure.

The observed statistic is sensitive to representation choices in a subset of recordings. N2-only thresholds changed T_{cs} by more than 20% in 12 of 38 recordings (factors 0.68 to 1.98); whether the test outcome would change cannot be assessed, because no surrogate distribution exists with these thresholds. Two of the four graph nodes are not EEG channels, so the states mix cortical, ocular and muscular activity.

4.3 What the pre-specified and post hoc results show

Pre-specified. The prespecified group criterion of an upper-tail excess was not met under either primary scheme. Neither sensitivity nor specificity was estimated against a known ground truth.

Post hoc. The magnitude of T_{cs} is uninformative on its own. Stored-replica exceedance fractions, not just surrogate means, show that many surrogate values exceed the fixed nominal χ^2_{36} benchmark (Table 7). These fractions are not calibrated false-positive rates. Exceeding this nominal reference therefore does not by itself identify nonlinear or higher-order dynamics of the signal; this is consistent with the result that functions of Markov processes need not be Markov [6]. Whether the asymptotic reference is also miscalibrated for these sparse tables would require a conditional test preserving each run's initial state and first-order transition counts [10, 45], which we did not perform.

Post hoc. The three nulls ask different questions: their z-scores were only partly concordant (ρ_s 0.65–0.89) and their tails behaved differently, so the choice of null determines what is being tested [13, 14].

4.4 Relation to other work

Tests of Markov order that preserve the lower-order structure of a symbol sequence exist: surrogate sequences that preserve the lower-order transition structure [9], an exact test that conditions on the observed transition counts [10] and constrained Markov-order surrogates [11]; CMI-based order estimation with randomization tests has been compared with asymptotic criteria [5]. Combined with raw-signal surrogates, or replaced by a conditional test preserving each run's initial state and first-order transition counts [10, 45], such tests would target second-order dependence directly instead of mixing it with lower-order properties that the signal surrogates do not preserve. Discrete sequences derived from EEG are often not first-order Markov: EEG microstate sequences show non-Markovian and non-stationary structure [2], and sleep-stage sequences depart from simple Markov dynamics, for example in the power-law distribution of the durations of wake episodes during sleep [46]. The states studied here are feature-defined states within N2, not sleep stages or microstates, so these results are not directly comparable. Non-stationarity is a known source of spurious rejections in surrogate testing [13, 17]. In this design, separate randomization preserved run boundaries and run-specific spectral constraints; preservation of symbolic occupancies and transition structure was not established. Stationarization within runs and redistribution of drift power remain candidate mechanisms, not demonstrated explanations.

4.5 Limitations

The study used one public dataset of 19 participants recorded in 1994 with two EEG derivations and scored by the R&K rules; the participants had mild sleep-onset complaints, every participant contributed one temazepam night, temazepam may alter N2 microstructure, and both condi-

tions were pooled in the combination step. The analysis is retrospective, its pre-specification is internal, the analysis recordings had been used before (Section 2.9), and one analyst carried out all steps. The recordings have now also been used for the post hoc analyses reported here, so any confirmatory test of these observations requires new data. No positive control has been validated; spectral preservation may retain state-sequence structure, but per-epoch symbol agreement has not been measured, so the test has unknown sensitivity to second-order dependence of the state sequence; limited sensitivity is a theoretical concern, not a measured power result (Section 4.2). The surrogates were generated from the raw signal of each run without detrending or end-matching, so drift and the recording filters as stored enter the run-level null and can bias it in either direction; the spectral audits verify the transform, not the preservation of the features. The exact rank-model reference assumes untied surrogate values and independence between nightly ranks and between participants; direct examination of all 114 recovered surrogate ensembles (227,886 values) found no repeated statistic values within an ensemble. This supports the untied-ensemble condition for these saved values, but does not establish exchangeability or independence between observed nights. The ensembles are recovered for review but have not yet been publicly deposited. The production orchestrator, the download receipt and the result archive are not yet public (the operator is available only from the repository history), so the production run cannot yet be reproduced independently. G2 has not been completed, and the mechanisms discussed in Section 4.2 are untested. The analysis does not address biological mechanisms, causality, treatment effects or clinical questions. An AI system assisted with code review, the post hoc analysis code and the drafting of this manuscript, including interpretive text such as the candidate mechanisms in Section 4.2 (Declarations).

4.6 Next steps

First document and supply the frozen contract and decision record, then close the open numerical and completion-label safeguards before any full G2 run. A targeted raw-signal diagnostic must generate new surrogates and explicitly save epoch features and symbols; stored thresholds can be inspected immediately, but scalar T^* values cannot reconstruct state agreement. An exploratory symbolic test should condition on each run's initial state and first-order transition counts [10, 45], with power assessed on known heterogeneous first-order processes and known alternatives. Full G2 is an execution diagnostic and cannot reopen G1. Confirmatory inference requires independent data and a plan frozen before their analysis.

5 Conclusions

Under the archived decision rule frozen before the production run, the group-level tests provided no evidence of an excess of the four-state N2 statistic under both primary common-phase schemes. This bounded negative result concerns four symbols derived from covariance-based EEG/EOG/EMG features and does not establish absence of second-order dependence, whose detection sensitivity has not been validated. The post hoc lower-tail pattern was concentrated in the 27 outputs already available when the diagnostic plan referred to it; the subsequent 11 recordings did not reproduce that tail pattern. Values of the observed size also arose in random-phase surrogates preserving the specified spectra, so the statistic's magnitude alone does not identify higher-order signal dynamics. The lower-tail shift under the run-level scheme is descriptive and post hoc. The frozen G1 result is retained, with G0 supported by the recovered stored execution evidence. Full G2 and a validated raw-signal positive control remain outstanding.

Declarations

Data availability. The raw data are available from PhysioNet (Sleep-EDF Database Expanded v1.0.0, <https://physionet.org/content/sleep-edfx/1.0.0/>) under the Open Data Commons Attribution License v1.0 (<https://opendatacommons.org/licenses/by/1.0/>) [19]. Supplementary Data S1 and S2 list the frozen recording-level outputs (114 rows) and the frozen group summary (3 rows) used in this paper. They are supplied with this manuscript package, under ODC-By 1.0, with a README that contains a column dictionary and the required notice. *Contains information from Sleep-EDF Database Expanded which is made available under the ODC Attribution License.* The full result archive of the production run (surrogate statistic values and median thresholds, observed epoch features and states, audits; SHA-256 20b6e95d...), the production analysis package (SHA-256 9ab675e3...) and the decision record BM-A028 are not yet public; they will be deposited with persistent identifiers before an archival release of the repository.

Code availability. The project repository <https://github.com/ventilos/brain-model> [35] is not an independently verified public release. The supplied tree 2f3976aa7ec7 is a tree identifier, not a public commit; the source snapshot predates this manuscript revision. Supplementary Code S1 reproduces the baseline S1/S2-derived calculations. Code S3 (posthoc.py) reproduces the exploratory 27/11 partition and equal-night rank-model case, records inputs and FFT diagnostics, and summarizes Data S3. Its ID-based partition encodes the previously reported split; it does not recover execution chronology independently. Code S2 reproduces Table 7 from the archived ensembles; Data S3 supplies all 114 fractions. The frozen R3 package and BM-A028 decision JSON and Markdown are attached in the internal revision package without modification. Complete reproduction also requires the original EDF data and archived execution records; the full result archive has not been publicly deposited. No DOI, public tag or remote release is asserted.

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Competing interests. The author declares no competing interests.

Ethics. Ethical approval and informed consent were not required for this secondary analysis of publicly available, de-identified data released under ODC-By 1.0; the original study is described in [18]. No new data were collected.

Use of AI tools. Claude (Anthropic; model identifier reported in the earlier manuscript as claude-opus-5-5) assisted code review, post hoc code, drafting and critical review in September–October 2026. OpenAI Codex in ChatGPT Work assisted audits, recovery and checking of stored evidence, arithmetic, interpretation, manuscript revisions 2.1–2.6, supplementary calculation scripts, and LaTeX PDF generation and visual verification. The supplied version-2.3 PDF includes OpenAI C2PA provenance reporting ChatGPT and gpt-6.1-sol-wm; this is file-generation metadata, not evidence of scientific correctness or an independent verification of every model used. The author supplied further AI-assisted reviews attributed to Grok and Claude; reported provenance is not independent human peer review. The author remains responsible for reviewing the scientific content and personal declarations before submission. This remains an internal author-review version, not a public deposit.

Author contributions (CRediT). Ł.W.: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing, Project administration.

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License. © 2026 Łukasz Westphal. This preprint is distributed under the Creative Commons Attribution 4.0 International License (<https://creativecommons.org/licenses/by/4.0/>). Supplementary Data S1 and S2: ODC-By 1.0; Supplementary Codes S1–S3: Apache License 2.0.

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A Inferential scope and methodological follow-up

This internal revision additionally incorporates the five-method audit of version 2.5: explicit package-completeness profiles, section-specific checks of the newer reported values, Code S3 and narrower wording about run preservation. It retains the preceding synchronization of code, bibliography, data documentation and manifests without changing the frozen results. Earlier revisions clarified tail direction, identified the run-level mean z , replaced an unsupported specificity claim, narrowed the measurement scope in the title and abstract, and distinguished stored-evidence G0 from a raw-signal reproduction. It adds the direct replica fractions in Table 7,

calculated from archive SHA-256 20b6e95d482c3371ce5b95ee44f861b6f39f327dbbd61d40fe276e674c6726f3. The 114 recording–scheme fractions, calculation script and input identity are supplied in the supplementary methodological package (Code S2 and Data S3). No frozen p-value, G1 threshold, production operator or Supplementary Data S1/S2 has changed.

The intersection–union rule is not invalid merely because the component nulls differ: it addresses excess under both references. Its size requires valid component tests, and its power against any specified alternative cannot exceed the power of either component. Neither the rule nor covariance preservation establishes zero power or inevitable failure for every alternative.

The archive stores replicate-level E and ILMI median thresholds, permitting an exploratory threshold-shift diagnostic without rereading EDF; these medians do not reconstruct individual surrogate features or symbols. The archive contains observed epoch features and states, permitting a separate exploratory symbolic-order analysis without rereading EDF. It does not contain the epoch-state sequences of each surrogate replica, so direct state agreement cannot be reconstructed from stored T^* values. A targeted raw-signal diagnostic would need to save those states and quantify feature changes, threshold crossings and state agreement. Such a diagnostic would remain post hoc and would not reopen G1.

A future symbolic reference must accommodate first-order heterogeneity between runs. Conditioning separately on each run’s initial state and first-order transition counts avoids estimating a separate kernel for each short run [10, 45]. Conditional sampling must be uniform, or use a justified exact Monte Carlo construction; the conditional sets of short runs may be degenerate and limit power. The conditional argument requires a time-homogeneous first-order transition kernel within each run; conditioning on its initial state does not require a stationary initial distribution. Within-run trends and signal-dependent stage boundaries can still confound an order interpretation. No such test has been implemented here. Participant-level resampling must preserve paired nights. Choices made on these observed data remain exploratory; independent confirmation requires new data and an earlier frozen plan.

Supplementary material

Supplementary Methods S1. Independent, untied rank-model reference for the Fisher statistic

Let $n = B = 1,999$. Under the specified untied rank model, with independent ranks between nights and between participants, the leave-one-out p-value of a randomly selected surrogate is U/n with U uniform on $\{1, \dots, n\}$. For a participant, $K = \min(U_1, U_2)$ has $P(K = k) = [(n - k + 1)^2 - (n - k)^2] / n^2$, and the contribution to F is $c(K) = -2 \ln \min\{1, 2K/n\}$, which is zero for $K \geq 1,000$, an event of probability $(1,000/1,999)^2 = 0.2503$. F is the sum of 19 independent copies of $c(K)$. Its distribution was evaluated on a grid of width 10^{-4} by fast Fourier transform convolution using rounding bounds [31], once with every contribution rounded down and once rounded up; the two results bracket the exact tail probability and differed by less than 4×10^{-5} at the observed statistics. The same calculation with $n = 2,000$, the grid of the observed plus-one p-values, gives the reference distribution of the observed statistic under this model; its tail probabilities at the observed statistics agree with those for $n = 1,999$ to within 1×10^{-5} . Its upper-tail probability is 0.0500 at the 95th percentile of the exact meta-null (48.98), and 0.0501, 0.0508 and 0.0505 at the frozen Monte Carlo 95th percentiles of the run-level, epochwise and independent-channel schemes (48.97, 48.90 and 48.93). The version 2.0 supplement reports that a Monte Carlo check with 10^7 draws (seed 987654321) gave 0.87681, 0.12718 and 0.11648 (standard error about 0.0001) at the three observed statistics, against rank-model values of 0.87679, 0.12706 and 0.11635; that large Monte Carlo run was not repeated for versions 2.1 or 2.2.

For a dependence counterexample with independent participants, nightly values ($U_s, 1 - U_s$) yield uniform Bonferroni participant p-values and a χ^2_{38} Fisher statistic. At the independent-night rank-model critical value 48.98, its tail is approximately 0.109, not 0.05. This illustrates a calibration limitation of that rank model, not an estimate of the ST44 pipeline error. Conversely, identical nightly ranks give the post hoc tails reported in Section 3.2. Both examples leave the archived decision unchanged.

Supplementary Data S1 and S2 and Code S1

Supplementary Data S1 (`BM_ST44_S1_recording_by_variant.csv`; SHA-256 78763c75160bfd8b3cb5d5e033a79d5b12f85f294f1ff4897c6c3051e073dabd) has one row per recording and surrogate scheme (114 rows) with the participant, night, condition, valid and N2 epoch counts, number of runs, n_C , df_{eff} , T_{cs} , CMI, CMI_{MM} , T_{v2} , T_{cs} with N2-only thresholds, and the surrogate mean, standard deviation, number of exceedances, p-value and z. All values are copied as text from the frozen R3 outputs. Supplementary Data S2 (`GROUP_SIGNAL_NULL_SUMMARY.csv`; SHA-256 048e1cd50067a5b6ed9d55610c19f3845b28069c8befa0e9bb04e029e16d44c1) is a byte-exact copy of the frozen group summary (3 rows). Both files are distributed under ODC-By 1.0 together with `README_DATA.txt`, which contains a column dictionary and the notice “*Contains information from Sleep-EDF Database Expanded which is made available under the ODC Attribution License.*” Supplementary Code S1 (Apache License 2.0) consists of `build_s1.py` (SHA-256 38fa43411c890272ff11d54ba41eee622d32655c7bcb4c8f07e0d2e5b37b9462), `analysis.py` (SHA-256 97fa1d09f6208f9f4b8c46351e7a9bec456670028dc349394e6977d81529f22a) and `mc_check.py` (SHA-256 e78c03731d4487acb6b1b355a9eaae68a680e4d3d548197d447bbe464ad6924a). `build_s1.py` documents how S1 was assembled from frozen outputs that are not yet public and checks that the values agree between the frozen source files (within 10^{-9} for T_{cs} and z, and 10^{-12} for p and the CMI values). `analysis.py` reports numerical residuals for the identities $p = (1 + \text{exceedances})/2,000$, $z = (T_{obs} - \bar{T}^*)/sd(T^*)$ and the CMI identities within floating-point precision, recomputes the Fisher statistics and participant p-values, computes the exact meta-null and produces the descriptive numbers, tables and figures already present in version 2.2; the replica fractions are reproduced by Supplementary Code S2 (`calculations/reproduce.py`), using the non-public result archive; Data S3 (`CHI2_SURROGATE_FRACTIONS.csv`) supplies the 114 fractions; `mc_check.py` is the Monte Carlo check of Supplementary Methods S1. Code S3 (`posthoc.py`) reproduces the 27/11 partition and equal-night rank-model case; `check_posthoc.py` binds the new reported values to their manuscript sections. FFT rounding brackets delimit grid discretization; numerical FFT error is monitored by mass checks, not rigorously bounded here.

Table S1: Per-recording structure and observed statistics. Part., participant; Cond., condition (pla, placebo; tem, temazepam); Valid, valid epochs; N2, N2 epochs; Runs, N2 runs; n_C , common-support positions; df_{eff} , descriptive support-count index; CMI and CMI_{MM} , plug-in and Miller-Madow-type conditional mutual information (nats); $T_{\text{cs}}^{\text{N2}}$, T_{cs} with N2-only thresholds; T_{v2} , legacy statistic.

Recording	Part.	Cond.	Valid	N2	Runs	n_C	df_{eff}	T_{cs}	CMI	CMI_{MM}	$T_{\text{cs}}^{\text{N2}}$	T_{v2}
ST7011	1	pla	1,092	529	82	385	34	43.0	0.056	0.012	64.8	201.4
ST7012	1	tem	1,035	586	64	476	36	63.1	0.066	0.028	91.5	177.7
ST7021	2	tem	919	412	39	338	35	63.2	0.094	0.042	71.9	161.6
ST7022	2	pla	942	353	52	263	34	40.9	0.078	0.013	40.0	146.5
ST7041	4	pla	993	536	30	481	35	79.0	0.082	0.046	83.9	151.6
ST7042	4	tem	1,129	622	38	553	35	72.5	0.066	0.034	88.7	161.8
ST7051	5	tem	1,014	368	48	291	31	40.1	0.069	0.016	41.3	113.5
ST7052	5	pla	1,029	396	25	346	31	56.4	0.081	0.037	63.8	117.9
ST7061	6	pla	1,008	589	32	534	33	80.0	0.075	0.044	75.7	131.5
ST7062	6	tem	952	581	33	522	35	70.2	0.067	0.034	92.0	133.5
ST7071	7	pla	816	201	40	132	19	58.5	0.222	0.150	54.5	126.3
ST7072	7	tem	908	345	44	279	25	76.4	0.137	0.092	104.1	134.6
ST7081	8	tem	964	468	58	377	34	56.6	0.075	0.030	61.0	138.3
ST7082	8	pla	930	450	44	375	34	42.8	0.057	0.012	30.5	123.1
ST7091	9	tem	943	464	45	388	28	85.8	0.111	0.074	86.1	159.7
ST7092	9	pla	923	354	52	267	35	51.8	0.097	0.032	43.1	142.4
ST7101	10	pla	1,042	303	57	205	32	28.9	0.071	-0.008	57.3	142.5
ST7102	10	tem	980	258	27	210	31	33.4	0.079	0.006	35.8	91.2
ST7111	11	tem	1,017	408	24	367	34	64.7	0.088	0.042	63.4	108.7
ST7112	11	pla	978	388	33	328	36	44.6	0.068	0.013	46.7	108.5
ST7121	12	pla	1,026	452	57	360	36	28.3	0.039	-0.011	34.7	126.0
ST7122	12	tem	945	480	21	441	36	48.2	0.055	0.014	55.1	95.5
ST7131	13	tem	859	470	27	421	36	54.7	0.065	0.022	53.5	110.9
ST7132	13	pla	852	384	39	318	33	51.9	0.082	0.030	49.8	122.5
ST7141	14	pla	858	364	34	304	29	88.9	0.146	0.098	83.6	149.5
ST7142	14	tem	873	401	26	355	33	43.9	0.062	0.015	63.9	98.7
ST7151	15	pla	890	304	53	213	32	57.4	0.135	0.060	47.2	156.2
ST7152	15	tem	1,041	447	40	378	31	67.4	0.089	0.048	62.9	133.6
ST7161	16	tem	1,057	445	46	361	34	43.1	0.060	0.013	44.8	138.5
ST7162	16	pla	988	430	50	337	25	61.3	0.091	0.054	75.8	158.5
ST7171	17	pla	961	294	47	218	28	46.1	0.106	0.041	52.5	121.5
ST7172	17	tem	1,000	382	52	294	28	87.3	0.148	0.101	59.5	188.1
ST7181	18	tem	959	569	30	511	32	84.0	0.082	0.051	75.0	162.1
ST7182	18	pla	967	525	35	463	35	78.9	0.085	0.047	65.0	148.4
ST7191	19	tem	955	423	34	360	32	61.6	0.085	0.041	64.5	133.4
ST7192	19	pla	931	545	26	502	36	57.9	0.058	0.022	75.3	102.3
ST7201	20	pla	921	610	41	539	36	119.4	0.111	0.077	101.9	200.1
ST7202	20	tem	897	625	47	542	35	74.1	0.068	0.036	82.5	164.8

Table S2: Per-recording surrogate results: surrogate mean $\overline{T^*}$ (SD), z and plus-one p ($B = 1,999$).

Recording	T_{obs}	Run-level (primary)			Epochwise (primary)			Independent-channel		
		$\overline{T^*}$ (SD)	z	p	$\overline{T^*}$ (SD)	z	p	$\overline{T^*}$ (SD)	z	p
ST7011	43.0	62.2 (12.7)	-1.52	0.9505	45.0 (9.5)	-0.22	0.5620	61.0 (12.3)	-1.47	0.9425
ST7012	63.1	72.3 (14.8)	-0.62	0.7200	67.0 (11.4)	-0.34	0.6235	72.7 (13.9)	-0.69	0.7425
ST7021	63.2	70.3 (14.9)	-0.47	0.6605	63.4 (11.7)	-0.02	0.4905	57.3 (12.8)	0.47	0.3125
ST7022	40.9	64.4 (13.2)	-1.78	0.9755	40.1 (9.0)	0.09	0.4380	56.5 (12.0)	-1.29	0.9110
ST7041	79.0	82.9 (15.5)	-0.25	0.5830	73.8 (12.1)	0.44	0.3290	76.8 (14.3)	0.15	0.4315
ST7042	72.5	64.9 (14.1)	0.54	0.2700	76.5 (13.2)	-0.30	0.6000	72.7 (14.7)	-0.02	0.4790
ST7051	40.1	71.1 (13.8)	-2.25	0.9955	50.0 (10.0)	-0.99	0.8330	60.1 (13.4)	-1.49	0.9430
ST7052	56.4	59.3 (12.5)	-0.24	0.5715	51.4 (10.3)	0.49	0.2925	54.5 (12.0)	0.16	0.4070
ST7061	80.0	79.5 (15.5)	0.03	0.4670	74.8 (12.9)	0.40	0.3350	71.8 (14.6)	0.56	0.2750
ST7062	70.2	70.8 (14.9)	-0.04	0.4865	61.6 (11.6)	0.74	0.2335	77.8 (15.9)	-0.48	0.6680
ST7071	58.5	45.9 (9.6)	1.32	0.1055	45.9 (9.1)	1.37	0.0990	24.6 (7.9)	4.30	0.0005
ST7072	76.4	74.0 (14.7)	0.17	0.4235	55.0 (9.1)	2.35	0.0135	49.9 (13.0)	2.03	0.0300
ST7081	56.6	84.6 (16.6)	-1.69	0.9665	55.5 (10.6)	0.10	0.4345	78.1 (15.2)	-1.42	0.9355
ST7082	42.8	57.7 (12.8)	-1.16	0.8800	36.5 (8.3)	0.76	0.2145	62.7 (13.2)	-1.50	0.9475
ST7091	85.8	93.6 (18.2)	-0.43	0.6380	83.8 (12.4)	0.16	0.4345	99.4 (18.6)	-0.73	0.7610
ST7092	51.8	48.6 (11.1)	0.29	0.3570	56.7 (11.0)	-0.44	0.6435	51.1 (10.7)	0.07	0.4445
ST7101	28.9	51.1 (10.5)	-2.12	0.9920	51.3 (9.8)	-2.28	0.9955	34.3 (9.9)	-0.54	0.6895
ST7102	33.4	66.7 (13.3)	-2.51	0.9970	50.7 (9.9)	-1.75	0.9685	46.2 (12.2)	-1.05	0.8475
ST7111	64.7	71.9 (14.3)	-0.51	0.6760	52.8 (10.8)	1.10	0.1365	73.8 (15.2)	-0.60	0.7115
ST7112	44.6	61.8 (13.5)	-1.27	0.9130	49.1 (9.9)	-0.45	0.6680	63.0 (13.0)	-1.42	0.9275
ST7121	28.3	45.1 (10.2)	-1.66	0.9620	38.6 (8.6)	-1.19	0.8905	47.1 (10.3)	-1.82	0.9815
ST7122	48.2	76.6 (15.1)	-1.88	0.9790	48.5 (9.9)	-0.03	0.4875	80.2 (15.4)	-2.08	0.9900
ST7131	54.7	76.9 (15.7)	-1.41	0.9315	61.1 (11.8)	-0.54	0.6970	82.5 (15.7)	-1.77	0.9765
ST7132	51.9	51.5 (11.4)	0.04	0.4580	43.2 (9.4)	0.93	0.1700	54.3 (11.8)	-0.20	0.5660
ST7141	88.9	63.2 (13.1)	1.96	0.0355	68.5 (12.3)	1.65	0.0570	63.0 (13.1)	1.98	0.0310
ST7142	43.9	113.9 (18.8)	-3.72	1.0000	61.9 (11.2)	-1.59	0.9530	76.8 (15.4)	-2.14	0.9885
ST7151	57.4	51.0 (10.7)	0.60	0.2685	53.4 (10.6)	0.38	0.3460	34.6 (9.5)	2.40	0.0155
ST7152	67.4	58.1 (12.5)	0.75	0.2275	53.9 (10.9)	1.24	0.1155	53.0 (12.3)	1.17	0.1165
ST7161	43.1	55.4 (12.2)	-1.01	0.8440	57.7 (10.5)	-1.40	0.9220	59.6 (12.8)	-1.30	0.9200
ST7162	61.3	69.5 (13.9)	-0.59	0.7105	60.6 (10.5)	0.07	0.4625	68.3 (13.4)	-0.53	0.6850
ST7171	46.1	55.9 (12.3)	-0.80	0.7735	50.8 (9.9)	-0.48	0.6670	49.8 (10.8)	-0.35	0.6095
ST7172	87.3	80.8 (15.2)	0.43	0.3105	58.5 (10.3)	2.78	0.0060	67.4 (12.9)	1.54	0.0695
ST7181	84.0	99.2 (18.8)	-0.81	0.7775	80.2 (12.8)	0.30	0.3650	97.9 (18.4)	-0.76	0.7735
ST7182	78.9	56.0 (12.5)	1.84	0.0440	59.5 (10.7)	1.81	0.0415	55.4 (11.9)	1.97	0.0305
ST7191	61.6	62.0 (12.8)	-0.03	0.4975	71.2 (12.6)	-0.77	0.7695	64.6 (13.1)	-0.23	0.5705
ST7192	57.9	74.7 (15.4)	-1.08	0.8665	56.3 (10.8)	0.15	0.4180	72.2 (14.7)	-0.97	0.8375
ST7201	119.4	89.7 (17.0)	1.75	0.0445	95.7 (14.7)	1.61	0.0595	87.1 (16.8)	1.93	0.0375
ST7202	74.1	72.7 (14.4)	0.10	0.4390	65.4 (12.1)	0.72	0.2240	81.2 (16.0)	-0.44	0.6450

Table S3: Production audits (AUDIT_12X) of the R3 run, with their verbatim names.

No.	Audit	What is checked	Result
01	locked inventory	38 recordings from 19 participants	pass
02	leakage lock	none of participants 21, 22 and 24 in the inventory	pass
03	paired nights	two recordings per participant	pass
04	condition mapping	two distinct condition labels (placebo, temazepam) per participant	pass
05	Drive receipt hashes	SHA-1 of every polysomnogram and hypnogram file equals the frozen download receipt	pass
06	ST7022 legacy regression	legacy statistic of ST7022 reproduces the frozen value 146.4779585335193 within 10^{-6}	pass
07	autospectrum preservation	largest absolute periodogram deviation divided by the largest periodogram value (all channels and frequencies; all epochs of the run for the epochwise scheme) below 10^{-10} ; first run of the first attempt per recording and scheme	pass (1.04×10^{-15})
08	common-phase cross-spectrum preservation	as 07 for the cross-periodograms of the two common-phase schemes	pass (9.98×10^{-16})
09	provenance-locked checkpoints	every recording checkpoint matches its contract (code and input hashes, B , seeds, schemes)	pass
10	surrogate acceptance rate	$B/\text{attempts} \geq 0.99$ for every recording and scheme	pass (min 1.000)
11	finite-ensemble pseudo-rank calibration	unit test of a re-implementation of the rank formula: $\#\{a \geq t\}/10$ equals $(10, 9, \dots, 1)/10$ for t in $a = (1, \dots, 10)$	pass
12	group calibration schema	columns, three schemes, statistic T_{cs} , 19 participants, B , M , group $p \in [1/(M+1), 1]$	pass

Table S4: Recordings with $p < 0.05$ under at least one scheme ($p < 0.05$ in bold; post hoc).

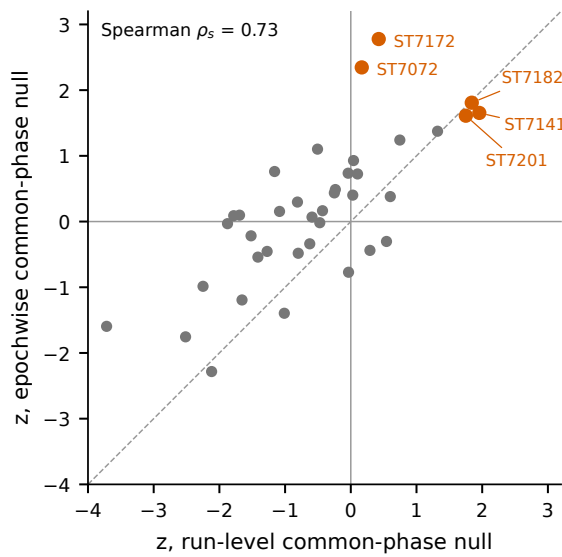
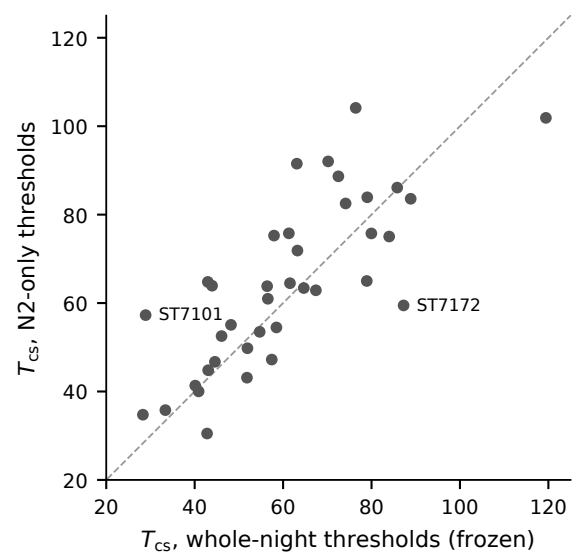
Recording	Participant	Condition	Recording-level p			Both primary ^a
			Run-level	Epochwise	Independent-channel	
ST7071	7	placebo	0.1055	0.0990	0.0005	no
ST7072	7	temazepam	0.4235	0.0135	0.0300	no
ST7141	14	placebo	0.0355	0.0570	0.0310	no
ST7151	15	placebo	0.2685	0.3460	0.0155	no
ST7172	17	temazepam	0.3105	0.0060	0.0695	no
ST7182	18	placebo	0.0440	0.0415	0.0305	yes
ST7201	20	placebo	0.0445	0.0595	0.0375	no

^a $p < 0.05$ under both primary schemes.**Table S5:** Temazepam minus placebo difference of z per participant ($n = 19$; post hoc; descriptive; not a treatment test).

Scheme	Mean difference	Median difference	Temazepam > placebo	Wilcoxon p
run-level	-0.57	-0.40	7/19	0.275
epochwise	-0.18	-0.12	8/19	0.541
independent-channel	-0.71	-0.77	6/19	0.080

Table S6: Integrity register of the artifacts referred to in this paper (SHA-256). The digests are also listed in TABLE_S6_SHA256.txt of the supplement.

Artifact	SHA-256	Availability
R3 analysis package (zip)	9ab675e3acd157b21845f5f242b43274337c6cb4f5121efc59ad03a84110414f	not yet public
operator BM-A004	e82a9fba5cba5892145a7d8401e15552a10efe2ef35654181551be9d04bd0080	repository history
orchestrator BM-A011	eefe0a421042b799df9b02811c355725d2940b8cdfe8e690472d1964f1182e21	not yet public
decision rules R3	1e9c2f1b2f1159d9d26668ea9bc13f195040ffa20e35c86ccc97c1c465cc2b14	not yet public
download receipt	0543a191c33b038f6e8dde32a0c8483892505a1bdf5460fe644a4b392009d629	not yet public
R3 result archive (zip)	20b6e95d482c3371ce5b95ee44f861b6f39f327dbbd61d40fe276e674c6726f3	not yet public
G0/G1 decision BM-A028 (JSON)	e37c36d089afd09ede83b1949b367552fa1d03c00530f4c0a5680deef17acde2	not yet public
Analysis plan BM-A027	f6403115755901861b8e202495841fb38d126a4c89ced2f80241ca569573fb90	repository
Amendment-001	ccf4a7b5fedacdf417fcb048c5639220a4116cc4e0c3c7168462d7cc8c31497e	repository
Amendment-002	20d4aa957f995182801e332fdf0ed00abadd079cb4366623695918f0b86d7a9f	repository
G2 runner BM-A031 Rev2	f1619b28939b6a68c812ca8cc9cd4ec7e00eb4aaff9924cc99b061e5da005d7b	repository
Version 1.0 of this paper (PDF)	3e70d1dbd3cb9d847a52d5ce0c28cbd13cad3800f9ed75de6241583588717f89	historical

**Figure S1:** Recording z-scores under the two primary schemes (post hoc). Orange, recordings with $p < 0.05$ under at least one primary scheme (only ST7182 under both); gray, other recordings; dashed line, identity.**Figure S2:** T_{cs} with the frozen whole-night thresholds against T_{cs} with N2-only thresholds (post hoc); dashed line, identity.